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Knowledge-aware modeling of group commonality for group recommendation

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ABSTRACT

Group Recommendation (GR) aims to offer recommendations that satisfy the entire group. Due to the inherent sparsity of group-item interactions in GR, relying solely on individual-level preference aggregation is often insufficient for producing high-quality recommendations. In contrast, mining group-level commonality that reflects shared behavioral patterns can help mitigate this challenge. Some existing methods attempt to model group commonality based on the number of overlapping users across groups. However, this approach often fails in sparse settings where shared users between groups are absent, leaving the data sparsity issue unresolved. To tackle these issues, we propose a novel model based on Knowledge-Aware Modeling of Group **Com**monality for Group **Rec**ommendation (ComRec). ComRec eliminates the reliance on overlapping users by modeling fine-grained commonality from the item side. Specifically, we construct a Group Collaborative Knowledge Graph (G-CKG) by integrating group members' interactions, membership relations, and item knowledge, enabling the capture of multi-hop relational paths for each member. We then extract fine-grained commonality by fusing multiple relational representations with an orthogonal constraint to ensure signal independence. A novel commonality attention mechanism further aggregates member entity representations to derive the overall group-level commonality representation. Beyond modeling group commonality, we further consider the specific group composition by introducing a user-based fine-tuning module that refines the group representation through member-level differences. The results show that our model significantly outperforms existing methods in terms of classification accuracy and interpretability on Yelp and MovieLens-20M datasets, while effectively addressing the data sparsity issue in GR.

1. Introduction

As social beings, humans naturally engage in group activities such as sharing meals, watching films, or traveling together, which gives rise to the task of Group Recommendation (GR). Due to the complexity of group decision-making and the diversity of preferences within a group, groups should not be treated as a special type of individual user with traditional recommendation methods (Amer-Yahia, Roy, Chawlat, Das, & Yu, 2009; Quintarelli, Rabosio, & Tanca, 2016). Instead, a dedicated algorithm is needed to simulate the group decision-making process and generate satisfactory recommendations. This necessity has led to the emergence of GR systems.

Early GR approaches typically rely on manually predefined strategies, such as averaging (Shin & Woo, 2009), to aggregate individual ratings into group ratings. These approaches are data-independent and lack dynamic flexibility. More recently, researchers have adopted data-driven deep learning models (Cao et al., 2018; Chen et al., 2022; Deng, Li, Liu, Ali, & Shao, 2021; Gan et al., 2025; Jia, Zhou, Dong, & Pan, 2021; Li, Wang, Lai, & Yuan, 2023; Sankar et al., 2020; Wang et al., 2020; Wu et al., 2023; Xu et al., 2024; Ye et al., 2024; Zhang et al., 2021; Zhao, Zhang, Bai, Nie, & Yuan, 2024) to model group preferences by aggregating individual preference embeddings. For example, AGREE (Cao et al., 2018) and MoSAN (Vinh Tran et al., 2019) dynamically aggregate member preferences through attention mechanisms, HyperGraph (Jia et al., 2021) and S^2-HHGR(Zhang et al., 2021) integrate member nodes using hypergraphs, and CubeRec (Chen et al., 2022) represents group preferences through the geometric structure of hypercubes.

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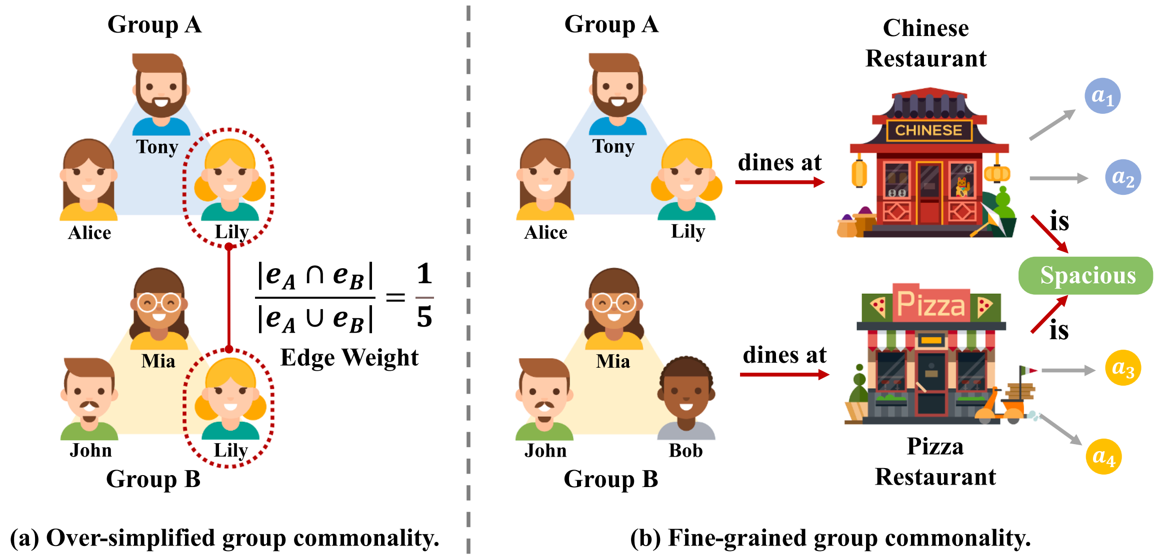


Fig. 1. (a) and (b) respectively illustrate the oversimplified group commonality represented solely by the number of shared users and the fine-grained group commonality derived from the KG.

Although these methods have achieved promising results, the prevalent sparsity of group-item interaction data in GR systems presents a big challenge. Approaches that focus solely on individual-level preference aggregation struggle to effectively extract information from limited interactions, which restricts their ability to generate high-quality recommendations. In this context, identifying group commonality that captures shared behavioral patterns becomes essential. While sparse interactions may not fully reflect each group's unique characteristics, they can still reveal transferable preference patterns across groups. For example, regardless of the group type, when dining together, they may all consider whether a restaurant has enough space to accommodate everyone. Similarly, when selecting a travel destination, transportation convenience is often prioritized to ensure that members can easily gather. It is worth noting that some methods (Wu et al., 2023; Xu et al., 2024) indirectly model group commonality through member overlap graphs to enhance group representation, where the number of shared members indicates the similarity between groups. As shown in Fig. 1(a), groups A and B are connected by an edge due to the shared member, Lily, with the edge weight calculated as the number of shared members over the total, which in this case is 1/5. However, this approach fails when there are no shared users between groups, and the 1/5 does not accurately represent their similarity. Group commonality, in fact, requires more fine-grained modeling from the item side, not just the number of common members. In Fig. 1(b), although groups A and B have no shared members and interact with two completely different restaurants, commonality information can still be propagated through the shared attribute 'Spacious' between the restaurants. This highlights that disentangling group commonality at a fine-grained level from the item side can more effectively capture similar preference patterns and alleviate data sparsity.

To achieve fine-grained modeling of group commonality from the item side, we introduce knowledge Graph (KG) into the GR task. KG, characterized by their structured and semantic nature, effectively represent entities and their complex relationships, offering a natural advantage for modeling group commonality. As shown in Fig. 1(b), a KG can flexibly represent restaurant attributes using triplets, such as (*Restaurant, is, Spacious*), and further utilize this information to extract higher-order group commonality. Unlike prior work, such as (Deng et al., 2021; Lin, Chen, & Huang, 2023), which uses the KG merely as auxiliary information to mitigate data sparsity in GR, this study goes further by fully exploiting the structural information within the KG to capture group commonality at a finer granularity. In addition to group commonality, we also consider group difference in the decision-making process, as different groups may exhibit varying choices due to the specific preferences of their members. As shown in Fig. 1(b), although both Group A and Group B care about the 'Spacious' attribute when selecting a restaurant, their differing member preferences lead Group A to choose Chinese food and Group B to choose pizza. Therefore, this study defines the group decision-making process by integrating group commonality and difference, emphasizing the dynamic relationships between different groups.

To this end, we propose a novel model based on Knowledge-Aware Modeling of Group **Com**monality for Group **Rec**ommendation (ComRec). To enable fine-grained modeling of group commonality from the item side, we construct a Group Collaborative Knowledge Graph (G-CKG) by integrating group member-item interactions, group memberships, and item knowledge. G-CKG enables the modeling of higher-order relational paths through a recursive multi-hop information propagation mechanism, laying a solid foundation for modeling both group commonality and difference. We then combine multiple relational representations from the G-CKG to capture fine-grained commonality, enforcing an orthogonal constraint to ensure distinct commonality signals are captured by each embedding. We then introduce an innovative commonality attention mechanism to aggregate group member entities, generating a unified group commonality representation. Finally, to account for the impact of different group compositions on the group representation, we introduce a user-based difference fine-tuning module that captures group-specific differences, facilitating the refinement of the group commonality representation. To evaluate the effectiveness of the proposed model, experiments are performed on three widely used benchmark datasets. The results demonstrate that ComRec consistently outperforms state-of-the-art methods in terms of Recall and NDCG. The main contributions of this work are presented as follows:

- We propose ComRec, a novel method that utilizes fine-grained group commonality mined from the G-CKG to effectively address data sparsity in GR.
- We define the group decision-making process by integrating both group commonality and difference, thereby enhancing the interpretability of GR.
- We introduce an innovative commonality attention mechanism that effectively aggregates group commonality representations and can seamlessly integrate with other models.
- We conduct experiments on three benchmark datasets to validate the effectiveness of our model. Experimental results have shown that our

model achieves superior performance compared to state-of-the-art baselines.

2. Related work

2.1. Group recommendation

In general, existing GR methods primarily aggregate individual preferences using various strategies to capture a group's overall interest (Gan et al., 2025; Wu et al., 2023; Zhao et al., 2024). From different aggregation perspectives, GR methods can be categorized into score aggregation and preference aggregation (Gan et al., 2025; Ye et al., 2024).

Score Aggregation methods first calculated item-wise preference scores for group members and then aggregated these scores using handcrafted predefined strategies to rank items at the group level. Three representative approaches were averaging (Shin & Woo, 2009), least misery (Amer-Yahia et al., 2009), and maximum pleasure (Boratto & Carta, 2011). The averaging strategy computed the mean rating across group members, while the least misery approach focused on maximizing the rating of the least satisfied member. In contrast, the maximum pleasure method prioritized options that maximized the satisfaction of group members. However, these rule-based methods were data-independent, lacked the flexibility for dynamic adjustment, and struggled to capture the complex interactions between group members.

Preference Aggregation methods, in contrast, employed data-driven strategies to learn and combine individual preference representations to generate group representations. To model the different weights of individual members in the group decision-making process, many attention-based methods (Cao et al., 2018, 2019a; Gan et al., 2025; He et al., 2024; He, Chow, & Zhang, 2020; Vinh Tran et al., 2019; Wang et al., 2020) were proposed. For instance, AGREE (Cao et al., 2018) directly integrated the attention mechanism with neural collaborative filtering, while MoSAN (Vinh Tran et al., 2019) extended this idea by employing a sub-attention network to compute the attention weights among group members.

However, the inherent sparsity of group-item interaction data in GR made it challenging to accurately learn user attention weights. Some methods addressed the data sparsity issue by incorporating social information (Alves et al., 2024; Guo et al., 2020; Ye et al., 2024, 2025; Yin et al., 2019). For example, SIGR (Yin et al., 2019) used a bipartite graph embedding model to learn and integrate both global and local social influences on users, while DisRec (Ye et al., 2025) decoupled user social influence from preferences and leveraged a social hypergraph to explore users' social influence both within and outside the group. Nevertheless, users' social roles became ineffective in newly formed groups. Other methods attempted to design various self-supervised learning objectives to alleviate the data sparsity issue (Chen et al., 2022; Li et al., 2023; Sankar et al., 2020; Wang, Liu, Zeng, Mu, & Li, 2024; Wu et al., 2023; Ye et al., 2025; Zhang et al., 2021). GroupIM (Sankar et al., 2020) maximized the mutual information between users and their respective groups. S^2-HHGR (Zhang et al., 2021) and SGGCF (Li et al., 2023) constructed different contrastive views by randomly removing user nodes and edges in the Hyper(Graph), while DHMAE (Zhao et al., 2024) removed data noise through a degree-sensitive hypergraph masking strategy. These methods generally followed a member completion paradigm, where data augmentation was achieved by masking or removing certain group members and reconstructing them. However, such operations could unpredictably disrupt the group structure-especially when critical individuals were mistakenly removed.

As discussed in Section 1, mining group commonality provided a new perspective for alleviating data sparsity. Some methods (Jia et al., 2021; Wu et al., 2023; Xu et al., 2024) constructed user overlap graphs based on the number of shared users, which facilitated the propagation of commonality information among similar groups. However, these methods have certain limitations. Firstly, they fail when there are no shared users between groups. Secondly, coarse-grained fuzzy modeling of group commonality results in a loss of information and may even introduce noise.

2.2. Knowledge graph-based recommendation

KGs were widely applied in recommendation systems due to their powerful ability to capture potential relationships and effectively address data sparsity. Generally, they can be classified into three categories: path-based methods, embedding-based methods, and GNN-based methods.

Path-based methods (Khan, Ma, Ullah, & Polat, 2022; Ma et al., 2019; Shi, Wang, Xing, & Xu, 2020; Zhao et al., 2020) identified potential recommendations by analyzing the paths or relationships between users and items. For example, CGAT (Liu et al., 2021) employs biased random walks over the KG to capture both short-range and long-range context around entities, while AMERec (Xie, Zheng, Chen, & Zheng, 2021) replaced traditional meta-paths with meta-graphs for feature extraction. However, these methods relied heavily on manually annotated domain knowledge, which limited the potential for enhancing recommendation performance and made them challenging to implement in real-world applications.

Embedding-based methods (Anelli, Di Noia, Di Sciascio, Ferrara, & Mancino, 2021; Gao, Peng, Lu, Claramunt, & Xu, 2023; Zhang, Yuan, Lian, Xie, & Ma, 2016; Zhang, Cai, Zhang, & Wang, 2020) utilized KG embedding algorithms to produce latent entity representations, which were subsequently fed into the recommendation framework. For example, DKN (Wang, Zhang, Xie, & Guo, 2018b) combined the textual content of news articles with TransD to obtain knowledge-level embedding representations, while KTUP (Cao, Wang, He, Hu, & Chua, 2019b) adopted TransH to both model user preferences and complete KG in a unified training process. However, these methods primarily captured only first-order relations, limiting their ability to encode long-range semantic connections and sequential dependencies along multi-hop paths.

GNN-based methods (Elahi et al., 2025; Jiang & Han, 2025; Li, Hou, & Li, 2024; Liu et al., 2023; Wang et al., 2021; Yuan et al., 2024) employed an iterative information propagation mechanism of graph neural networks (GNNs) to aggregate multi-hop neighbor information in the KG, effectively overcoming the limitations of the aforementioned approaches. Representative models include RippleNet (Wang et al., 2018a), which pioneered the introduction of preference propagation into KG; KGCN (Wang, Zhao, Xie, Li, & Guo, 2019a), which extended the GCN approach to KG-based recommendations; and KGAT (Wang, He, Cao, Liu, & Chua, 2019b), which incorporated an attention mechanism over the KG to refine recommendation signals.

Due to the more severe data sparsity issues in GR scenarios compared to individual recommendation settings, a few recent works (Deng et al., 2021; Jiang et al., 2023; Lin et al., 2023) integrated KG with GR systems. For example, KGAG (Deng et al., 2021) used KG as auxiliary information to obtain enhanced user knowledge representations, while Lin et al. (2023) treated both users and items as attributes of the group to obtain enhanced group knowledge representations. However, these methods did not fully harness the potential of KG in GR scenarios. In this approach, KG serves not only to mitigate the data sparsity problem in GR, but also to deeply explore group commonality information.

In summary, we compare our work with existing studies from six key aspects: (1) Data-Driven; (2) Attention Mechanism; (3) GNN-Based (4) Alleviating Sparsity; (5) Knowledge Graph; and (6) Modeling Commonality. The detailed comparison is shown in Table 1. The proposed ComRec integrates G-CKG into GR to extract fine-grained group commonality information. It overcomes the limitations of previous methods that rely on overlapping users to represent group commonality at a coarse level. Furthermore, ComRec provides a novel solution to the data sparsity issue in GR. Instead of directly aggregating individual preferences as most existing methods do, ComRec introduces an innovative commonality attention mechanism to effectively aggregate group commonality representations.

Table 1
Comparison of similarities and differences between ComRec and existing methods.

Paper	Data-Driven	Attention Mechanism	GNN-Based	Alleviating Sparsity	Knowledge Graph	Modeling Commonality
Shin and Woo (2009)	✗	✗	✗	✗	✗	✗
Amer-Yahia et al. (2009)	✗	✗	✗	✗	✗	✗
Boratto and Carta (2011)	✗	✗	✗	✗	✗	✗
Cao et al. (2018)	✓	✓	✗	✗	✗	✗
Cao et al. (2019a)	✓	✓	✗	✗	✗	✗
Vinh Tran et al. (2019)	✓	✓	✗	✗	✗	✗
Yin et al. (2019)	✓	✓	✗	✓	✗	✗
He et al. (2020)	✓	✓	✗	✗	✗	✗
Wang et al. (2020)	✓	✓	✓	✗	✗	✗
Guo et al. (2020)	✓	✓	✗	✓	✗	✗
Sankar et al. (2020)	✓	✓	✗	✓	✗	✗
Zhang et al. (2021)	✓	✓	✓	✓	✗	✗
Deng et al. (2021)	✓	✓	✓	✗	✓	✗
Jia et al. (2021)	✓	✓	✓	✓	✗	✓
Chen et al. (2022)	✓	✓	✗	✓	✗	✗
Li et al. (2023)	✓	✗	✓	✓	✗	✗
Jiang, Sun, Chen, and He (2023)	✓	✓	✓	✗	✓	✗
Lin et al. (2023)	✓	✓	✓	✗	✓	✗
Wu et al. (2023)	✓	✓	✓	✓	✗	✓
He et al. (2024)	✓	✗	✓	✗	✗	✗
Ye et al. (2024)	✓	✓	✗	✓	✗	✗
Zhao et al. (2024)	✓	✓	✓	✓	✗	✗
Wang et al. (2024)	✓	✓	✓	✓	✗	✓
Gan et al. (2025)	✓	✓	✗	✗	✗	✗
Ye et al. (2025)	✓	✓	✓	✓	✗	✓
Our Paper	✓	✓	✓	✓	✓	✓

3. Preliminaries

3.1. Knowledge graph

KG stores structured real-world information in the form of a heterogeneous graph, including entity relationships, item attributes, and more (Wang et al., 2019b, 2021). Let $\mathcal{V}$ be the set of real-world entities and $\mathcal{R}$ be the set of relationships. The KG $G_K = \{(h, r, t) | h, t \in \mathcal{V}, r \in \mathcal{R}\}$ consists of triplets, where each triplet (h, r, t) represents a relation r from the head entity h to the tail entity t. For instance, (*Robert John Downey, Portrayed, Iron Man*) denotes that *Robert John Downey* is the actor of the movie *Iron Man*. Note that $\mathcal{R}$ includes both original and inverse relations (e.g., *Portrayed* and *PlayedBy*). By linking items to their corresponding KG entities ($\mathcal{I} \subset \mathcal{V}$), the graph can characterize items with additional contextual information.

3.2. Task definition

Let $\mathcal{U} = \{u_1, u_2, \ldots, u_M\}$, $\mathcal{I} = \{i_1, i_2, \ldots, i_N\}$ and $\mathcal{G} = \{g_1, g_2, \ldots, g_K\}$ represent the sets of M users, N items and K groups, respectively. We consider two interaction types-user-item and group-item-which are recorded in the binary matrices $\mathbf{Y}_U \in \mathbb{R}^{M\times N}$ and $\mathbf{Y}_G \in \mathbb{R}^{K\times N}$. Specifically,

$$\mathbf{Y}_U(i,j) = \begin{cases} 1, & \text{if user } u_i \text{ has interacted with item } i_j, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

and $\mathbf{Y}_G$ is defined analogously for groups. Each group $g_t \in \mathcal{G}$ comprises a subset of users $\mathcal{G}_t = \{u_{t,1}, u_{t,2}, \ldots, u_{t,|\mathcal{G}_t|}\}$, where $|\mathcal{G}_t|$ is its size. Given a target group g_t, our our aim is to generate a ranked list of items that best match the collective preference of $\mathcal{G}_t$, formally defined as:

Input: User set $\mathcal{U}$, item set $\mathcal{I}$, group set $\mathcal{G}$, User-item interaction matrix $\mathbf{Y}_U$, Group-item interaction matrix $\mathbf{Y}_G$ and a KG G_K.

Output: A scoring function $f_{g_t} : \mathcal{V} \to \mathbb{R}$ that assigns a real-valued relevance score to each item for the target group g_t.

The key notations used in this study are summarized in Table 2.

Table 2
Notations and their definitions.

Notation	Description
u	user
i	item
g	group
$\mathcal{U}$	Set of all users
$\mathcal{I}$	Set of all items
$\mathcal{G}$	Set of all groups
$\mathcal{V}'$	Set of all entities
$\mathcal{R}'$	Set of all relations
$\mathcal{C}$	Set of all commonalities
$\mathbf{Y}_U$	User-item interactions matrix
$\mathbf{Y}_G$	Group-item interaction matrix
$\mathbf{e}_i$	Embedding of entity i
$\mathcal{N}_i$	Set of all neighboring nodes of entity i
ζ	Attention weight controlling relation representation
η	Attention weight controlling group commonality representation
W	Trainable weight
b	Trainable bias
α	Trainable weight balancing group commonality and difference
β	Trainable weight balancing group commonality and difference
d	Hyperparameter controlling latent dimension
$\lvert\mathcal{C}\rvert$	Hyperparameter controlling the number of commonalities
λ_1	Hyperparameter controlling orthogonality loss
λ_2	Hyperparameter controlling L_2 regularization loss

4. Methodology

In this section, we present a detailed description of the ComRec model. As depicted in Fig. 2, the overall architecture is composed of three major modules: (1) ***Knowledge Graph-based Information Aggregation***, which first constructs a G-CKG and then employs a graph neural network to infuse prior knowledge into its nodes, generating representations for users, items, and their relationships; (2) ***Group-based Commonality Modeling***,which integrates the relationship representations from (1) to obtain fine-grained commonality embeddings and ensures their independence through an orthogonal constraint. It then aggregates the group's member entities based on these embeddings using a commonality attention mechanism to derive the group's commonality

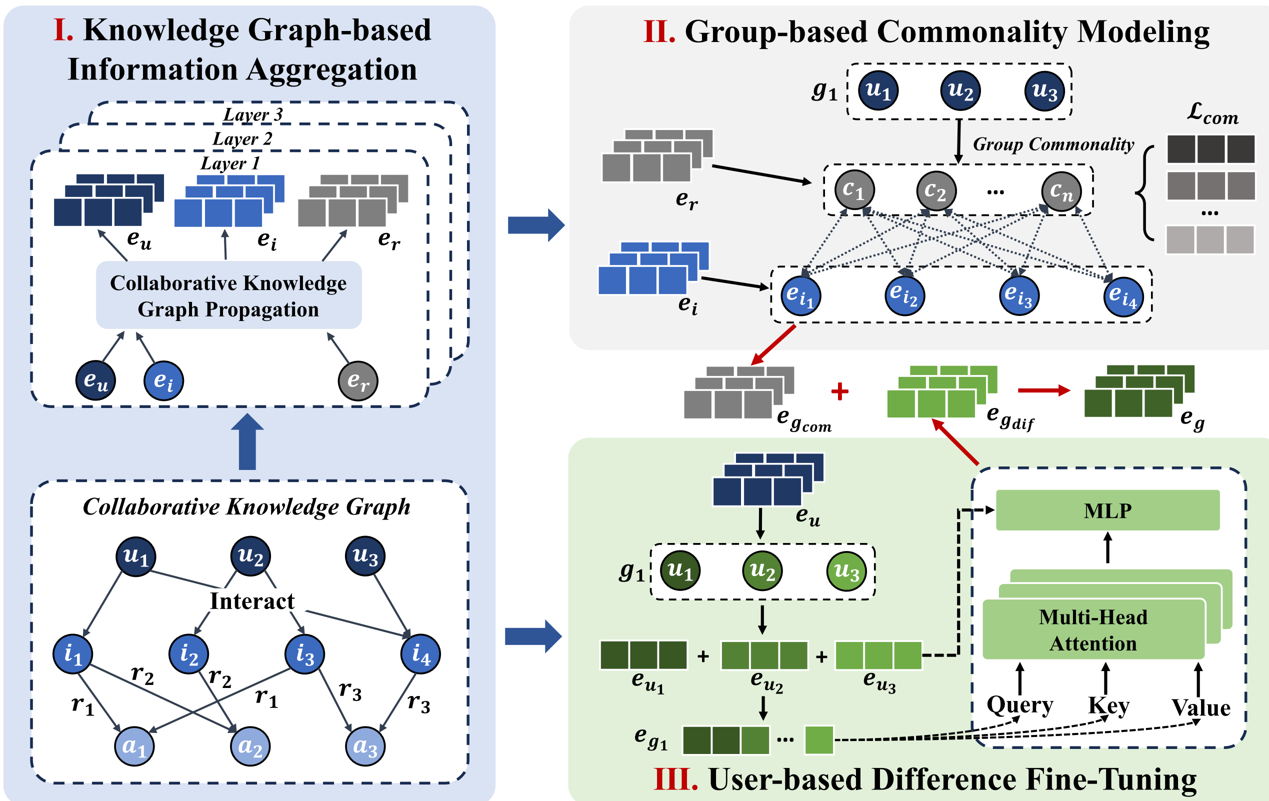


Fig. 2. Framework of knowledge-aware modeling of group commonality for group recommendation (ComRec), where u, i, a, and r respectively denote user, item, attribute, and relation nodes, and e denotes the embedding of nodes.

representation; (3) ***User-based Difference Fine-Tuning***, which employs a multi-head attention mechanism to aggregate higher-order user information across various interaction spaces, thus obtaining the group's difference representation. Finally, we combine the group commonality representation from (2) with the group difference representation from (3) to obtain the final group representation.

4.1. Knowledge graph-based information aggregation

In this subsection, we begin by detailing the construction of the G-CKG, followed by the recursive aggregation of entity representations and the transformation of multi-hop neighbors on the G-CKG.

4.1.1. Group construction of collaborative knowledge graph

Inspired by Li et al. (2024); Wang et al. (2019b), we construct a G-CKG based on the KG, which unifies user behaviors, group memberships, and item-related knowledge into a heterogeneous relational graph. As illustrated in the bottom-left corner of Fig. 2, each interaction pair in $\mathbf{Y}_U$ is modeled as a triplet $(u, interact, i)$ where $\mathbf{Y}_U(i,j) = 1$, and any two members $u_1, u_2 \in \mathcal{G}t$ are connected via a membership triplet $(u_1, isWith, u_2)$. In this way, user-item interactions and group memberships are seamlessly incorporated into the original KG G_K by introducing two new relation types: interact and isWith. The G-CKG is formalized as:

$$G_{CK} = \left\{(h,r,t) | h,t \in \mathcal{V}', r \in \mathcal{R}'\right\}, \tag{2}$$

where $\mathcal{V}' = \mathcal{V} \cup \mathcal{U}$ and $\mathcal{R}' = \mathcal{R} \cup \{interact, isWith\}$. This extension not only reveals entity attributes and relationships but also captures the preference characteristics of group members.

4.1.2. Propagation of collaborative knowledge graph

After constructing the G-CKG, we focus on learning representations for its entities and relations. Embedding-based methods to parameterize KG entities and relations have been widely considered in existing approaches (Lin et al., 2023; Wang et al., 2019b). However, in the context of GR, capturing group commonality necessitates high-order relational paths, such as $group \xrightarrow{r_1} user \xrightarrow{r_2} item \xrightarrow{r_3} attribute$. To achieve this, we leverage a GNN-based framework that incorporates a multi-hop neighborhood information propagation and aggregation mechanism. Since an item can be interacted with by multiple users, and a user can interact with multiple items, we can treat other entities connected to an entity as its attributes to capture the collaborative information between them. For instance, an item may be favored by a specific group of users, and vice versa. Let $\mathcal{N}_i$ define the set of attributes and first-order relationships for entity i, given by:

$$\mathcal{N}_i = \{(r,t) \mid (i,r,t) \in G_{CK}\}. \tag{3}$$

The relation-aware information from the connected entities is then integrated to compute the representation of entity i as follows:

$$\mathbf{e}_i^{(1)} = \frac{1}{|\mathcal{N}_i|} \sum_{(r,t)\in\mathcal{N}_i} \mathbf{e}_r \cdot \mathbf{e}_t^{(0)}, \tag{4}$$

where $\mathbf{e}_i^{(1)} \in \mathbb{R}^d$ represents the aggregated information from the first-order connections, d denotes the latent dimensionality, and $\mathbf{e}_t^{(0)}$ refers to the entity embedding of t. The dot product operation is used to incorporate the relation $\mathbf{e}_r$ into the entity $\mathbf{e}_i$'s embedding in order to distinguish cases of 'reaching the same node under different relations,' ensuring that the outcomes carry distinct meanings in various relational contexts. To propagate information through higher-order connections, we extend it to neighbors across multiple layers as follows:

$$\mathbf{e}_i^{(l)} = \frac{1}{|\mathcal{N}_i^{(l-1)}|} \sum_{(r,t)\in\mathcal{N}_i^{(l-1)}} \mathbf{e}_r \cdot \mathbf{e}_t^{(l-1)}, \tag{5}$$

where $\mathcal{N}_i^{(l-1)}$ refers to the set of entities that are directly connected to $\mathbf{e}_i^{(l)}$ at the $(l-1)$-th order. By propagating information through multiple layers of relational edges, entity representations are obtained at each layer and subsequently combined to form the final entity representations:

$$\hat{\mathbf{e}}_i = \mathbf{e}_i^{(0)} + \mathbf{e}_i^{(1)} + \cdots + \mathbf{e}_i^{(l)}. \tag{6}$$

4.2. Group-based commonality modeling

In this subsection, we first obtain diverse commonality representations by integrating relation embeddings, then apply an orthogonal constraint to ensure their independence, and finally use the commonality attention mechanism to aggregate group member entities and generate the group's commonality representation.

4.2.1. Representation of commonality

We define commonality as the factors a group collectively considers during decision-making, capturing shared behavioral patterns across groups. Different commonalities abstract distinct group behaviors, motivating us to model the group-item relationship at the commonality level. As shown in the upper right corner of Fig. 2, each group-item interaction pair (g, i) in $\mathbf{Y}_G$ where $(\mathbf{Y}_G(i, j) = 1)$ is decomposed into $\{(g, c, i) \mid c \in \mathcal{C}\}$, where $\mathcal{C}$ represents the set of commonalities shared by all groups. The size of $\mathcal{C}$, denoted as $|\mathcal{C}|$, is controlled by a hyperparameter. Inspired by Wang et al. (2021), we represent these commonalities by integrating different relation representations in the G-CKG rather than introducing new latent vectors. Formally, For each $c \in \mathcal{C}$, its representation is given by:

$$\mathbf{e}_c = \sum_{r \in \mathcal{R}'} \zeta(r, c)\mathbf{e}_r, \tag{7}$$

$$\zeta(r, c) = \frac{\exp(w_{rc})}{\sum_{r' \in \mathcal{R}'} \exp(w_{r'c})}, \tag{8}$$

where $\mathbf{e}_r$ denotes the entity embedding for relation r, $\zeta(r, c)$ is the attention score of r, and w_{rc} is a trainable parameter associated with both the relation r and the commanliy c. To ensure that different commonality representations capture unique and non-redundant information, we introduce an orthogonality constraint that minimizes the inner product between commonality vectors. Given $|\mathcal{C}|$ commonality representations $\{\mathbf{e}_{c_1}, \mathbf{e}_{c_2}, \ldots, \mathbf{e}_{c_{|\mathcal{C}|}}\}$, the loss function for the orthogonality constraint is formulated as:

$$\mathcal{L}_{\text{orth}} = \sum_{i<j} \left(\mathbf{e}_i^\top \mathbf{e}_j\right)^2. \tag{9}$$

Optimizing this loss promotes the orthogonality between different group commonalities, giving them clearer boundaries and thereby enhancing the interpretability of group behaviors.

4.2.2. Commonality attention mechanism

After obtaining the commonality representations $\{\mathbf{e}_{c_1}, \mathbf{e}_{c_1}, \ldots, \mathbf{e}_{c_{|\mathcal{C}|}}\}$ shared across groups, we further construct a unique commonality representation for each group g based on its member composition and the historical items it has interacted with, thereby generating personalized recommendations for the group. Considering a group g, let

$$\mathcal{N}_g = \{(c, p) \mid (g, c, p) \in \mathcal{C}\} \tag{10}$$

denote the interaction history and group members of group g. The commonality-aware information from both historical items and group members is then integrated to generate the group g's representation as follows:

$$\mathbf{e}_{g_{com}}^{(1)} = \frac{1}{|\mathcal{N}_g|} \sum_{(c,i) \in \mathcal{N}_g} \mathbf{e}_c \cdot \mathbf{e}_i^{(0)}, \tag{11}$$

where $\mathbf{e}_i^{(0)}$ is the ID embedding of first-order entity i. Through this, we obtain the group commonality fusion representation $\mathbf{e}_{g_{\text{com}}}^{(1)}$. However, for a given group, different commonalities have varying impacts. For instance, in the case of a family, the privacy of a restaurant may be more important than its affordability. To address this, $\eta(g, c)$ is introduced to quantify the contribution of commonality c to group g. Accordingly, we update Eq. (11) as follows:

$$\mathbf{e}_{g_{com}}^{(1)} = \frac{1}{|\mathcal{N}_g|} \sum_{(c,i) \in \mathcal{N}_g} \eta(g, c)\mathbf{e}_c \cdot \mathbf{e}_i^{(0)}, \tag{12}$$

$$\eta(g, c) = \frac{\exp(\mathbf{e}_c^\top \mathbf{e}_{g_{com}}^{(0)})}{\sum_{c' \in \mathcal{C}} \exp(\mathbf{e}_{c'}^\top \mathbf{e}_{g_{com}}^{(0)})}, \tag{13}$$

where $\mathbf{e}_{g_{com}}^{(0)} \in \mathbb{R}^d$ is the commonality embedding of group g, which helps personalize the importance score. After defining the first-order representation, the group commonality representation at the l-th layer is recursively generated to extract more structured information from higher-order neighbors, as follows:

$$\mathbf{e}_{g_{com}}^{(l)} = \frac{1}{|\mathcal{N}_g^{(l-1)}|} \sum_{(c,i) \in \mathcal{N}_g^{(l-1)}} \eta(g, c)\mathbf{e}_c \cdot \mathbf{e}_i^{(l-1)}, \tag{14}$$

where $\mathbf{e}_{g_{com}}^{(l)}$ is the representation of the group commonality representation at layer l. After propagating through l layers, the commonality representations of group g from each layer are aggregated by summation to form the final representation:

$$\hat{\mathbf{e}}_{g_{com}} = \mathbf{e}_{g_{com}}^{(0)} + \mathbf{e}_{g_{com}}^{(1)} + \cdots + \mathbf{e}_{g_{com}}^{(l)}. \tag{15}$$

4.3. User-based difference fine-tuning

Group commonality reflects the shared concerns and decision-making trends among groups. However, the individual preferences among group members introduce diversity into the decision-making process. To address this, we define group difference as the fine-tuning of group commonality based on the personal preferences of its members, allowing for a better capture of the unique and differentiated needs of different groups in decision-making. For a group $\mathcal{G}_t = \left\{u_1, u_2, \ldots, u_{|\mathcal{G}_t|}\right\}$, we first concatenate all of its members as follows:

$$\mathbf{C} = [\hat{\mathbf{e}}_{u_1}; \hat{\mathbf{e}}_{u_2}; \ldots; \hat{\mathbf{e}}_{u_{|\mathcal{G}_t|}}], \tag{16}$$

where $\mathbf{C}$ is the concatenation of users in group $\mathcal{G}_t$, and $\hat{\mathbf{e}}_{u_i}$ is the sum of each layer of user u_i obtained through recursion, as described in Eq. (6). To improve the model's capacity for capturing diverse subspace representations of users implied by different relational connections and to effectively capture the complex and diverse user node features in the G-CKG, the ComRec model employs a multi-head attention mechanism (Vaswani et al., 2017). This mechanism originates from the user node and extracts latent information from its neighboring nodes. The information in the i-th subspace is formulated as follows:

$$H_i = \delta\left(\frac{(\mathbf{C}W_i^Q)(\mathbf{C}W_i^K)^\top}{\sqrt{d}}\right)(\mathbf{C}W_i^V), \tag{17}$$

where the weight matrices $W_i^Q \in \mathbb{R}^{d\times d}$, $W_i^K \in \mathbb{R}^{d\times d}$, and $W_i^V \in \mathbb{R}^{d\times d}$ correspond to query, key, and value vectors, respectively, for the i-th subspace. Here, $\delta(\cdot)$ is the softmax function, and $\sqrt{d}$ serves as a normalization factor to mitigate overly large values in the inner product computation. We then concatenate the representations from different subspaces and project them to derive the final multi-head attention representation:

$$\mathbf{M} = [H_1; H_2; \ldots; H_h]W^O, \tag{18}$$

where h is a hyperparameter that controls the number of attention heads, and $W^O \in \mathbb{R}^{hd\times d}$ is the projection matrix. Finally, we concatenate $\mathbf{M}$ with the original user set $\mathbf{C}$, passing the combined result through the MLP layer to derive the final group difference representation:

$$\hat{\mathbf{e}}_{g_{dif}} = \text{ReLU}([\mathbf{C}; \mathbf{M}]W_1 + b_1)W_2 + b_2, \tag{19}$$

where $W_1 \in \mathbb{R}^{d\times d}$ and $W_2 \in \mathbb{R}^{d\times d}$ are trainable projection matrices, and $b_1 \in \mathbb{R}^d$ and $b_2 \in \mathbb{R}^d$ are bias vectors.

4.4. Model optimization

After obtaining the consensus and difference representations of the group, we use two parameters, α and β, to quantify their relative importance and enhance interpretability. These parameters are used to assign values to $\hat{\mathbf{e}}_{g_{com}}$ and $\hat{\mathbf{e}}_{g_{dif}}$, and the final group representation is obtained by summing them as follows:

$$\mathbf{e}_g = \alpha \hat{\mathbf{e}}_{g_{com}} + \beta \hat{\mathbf{e}}_{g_{dif}}. \tag{20}$$

As the task involves recommending top-K items for groups, we adopt Bayesian Personalized Ranking (BPR) (Rendle, Freudenthaler, Gantner, & Schmidt-Thieme, 2009)-a pairwise optimization strategy that encourages higher scores for observed items over unobserved ones. To this end, we refine the group preference representations by minimizing the group-item BPR loss, denoted as $\mathcal{L}_{group}$, formulated as:

$$\mathcal{L}_{group} = -\sum_{(g,i,i')\in\mathcal{Z}} \ln \sigma\left(\hat{y}_{g,i} - \hat{y}_{g,i'}\right), \tag{21}$$

where $\mathcal{Z}$ denotes the set of group training instances, and each instance (g, i, i') indicates that group g has interacted with item i but not with item i'. The function $\sigma(\cdot)$ represents the sigmoid activation, and $\hat{y}g, i$ and $\hat{y}g, i'$ are the predicted scores for items i and i', respectively, computed as:

$$\hat{y}_{g,i} = \mathbf{e}_g^T \mathbf{e}_i. \tag{22}$$

Finally, we jointly minimize the orthogonality loss $\mathcal{L}_{orth}$ Eq. (9) and the group preference loss to learn the model parameters by optimizing the following objective function:

$$\mathcal{L} = \mathcal{L}_{group} + \lambda_1 \mathcal{L}_{orth} + \lambda_2 \|\Theta\|_2^2, \tag{23}$$

where $\Theta = \{\mathbf{E}, \mathbf{W}, b_1, b_2\}$ denotes the model's parameter set, where $\mathbf{E}$ is the embedding table for all entities and relations, and $\mathbf{W}$ is the embedding table for all projection matrices. Additionally, λ_1 and λ_2 are hyperparameters that control the orthogonality loss and L_2 regularization term, respectively.

5. Experiments

In this section, we aim to answer the following research questions (RQs):

- **RQ1**: How does our model perform compared to state of-the-art approaches?
- **RQ2**: How do the major components of ComRec contribute to its overall recommendation performance?
- **RQ3**: Is ComRec's fine-grained modeling of group commonality based on the G-CKG more effective than existing approaches (e.g., overlap graphs), and is it transferable to other methods?
- **RQ4**: How do the hyper-parameter settings influence the performance of ComRec?
- **RQ5**: How does ComRec handle different numbers of group-item interactions, especially in sparse interaction settings?
- **RQ6**: Can ComRec effectively capture group commonalities and offer an intuitive and interpretable explanation?

5.1. Datasets

In GR scenarios, datasets typically include at least three types of information: group-item interactions, user-item interactions, and group membership. In this study, we perform comprehensive experiments on three widely used GR datasets. Below, we offer a detailed description of these datasets and the process of constructing the KG for each.

The first dataset, **Yelp**,[1] is a real-world dataset comprising approximately 160,000 businesses across eight major metropolitan areas.

Table 3
Dataset statistics.

Dataset	Yelp	MovieLens-Rand	MovieLens-Simi
# Users	35,665	3823	3823
# Items	29,242	22,349	10,364
# Groups	28,992	49,402	28,426
# U-I interactions	548,875	649,504	431,371
# G-I interactions	40,394	225,545	142,130
Avg. # items per user	18.93	169.89	112.84
Avg. # items per group	1.38	2.97	5.22
Avg. group size	2.95	8	5
# entities	132,705	152,569	124,108
# relations	42	39	32
# triplets	2,007,619	1,276,449	833,365

It primarily contains business-related information, including reviews, check-ins, and attributes. To ensure data quality, users and items with fewer than five interactions are removed before data processing. Following the methods described in Yin et al. (2019), Sankar et al. (2020), Deng et al. (2021), Chen et al. (2022), Benouaret and Tan (2023) group interactions are formed by integrating check-in data with social network information. Specifically, a group interaction is identified when a group of friends, according to the social network, simultaneously check into the same business within a 15-min window. We construct a KG using business-related information such as attributes, geographic details, and product classifications.

The other two datasets, MovieLens-Rand and MovieLens-Simi, are obtained from the **MovieLens-20M**.[2] The MovieLens-20M dataset contains approximately 20 million ratings and 465,000 tag applications generated by 138,000 users for 27,000 movies. Following the approches in Baltrunas, Makcinskas, and Ricci (2010), Vinh Tran et al. (2019), Deng et al. (2021), Zhao et al. (2024), Abolghasemi, Viedma, Engelstad, Djenouri, and Yazidi (2024), Zhang, Liu, and Feng (2025), we divide the groups into two categories, *random* and *similar*, according to the rating similarity among group members, and generate the corresponding MovieLens-Rand and MovieLens-Simi datasets.

In the MovieLens-Simi dataset, group members are selected based on shared interests, measured using Pearson Correlation Coefficient (PCC) (Cohen et al., 2009). Following the recommendation in Deng et al. (2021), we apply a similarity threshold of 0.27 to identify members with aligned preferences. In contrast, the MovieLens-Rand dataset consists of groups formed randomly, without enforcing any similarity among members. Consistent with previous studies (Deng et al., 2021; Yin et al., 2019; Zhou, Huang, Wang, & Chen, 2023), an item is considered a ground-truth item if all group members have assigned it a rating greater than 4. The dataset construction process is detailed in Algorithm 1. For constructing the KG, we adopt the method proposed in Deng et al. (2021); Wang et al. (2019a), where items are aligned with entities in Microsoft Satori by matching their names to the tail entities in triplets. Items that correspond to multiple or no entities are excluded to ensure mapping accuracy.

Table 3 summarizes the main statistics of the three datasets along with the corresponding KG details. Each dataset is randomly divided into a training set (80 %) and a test set (20 %). Additionally, 10 % of the training data is randomly sampled as a validation set for hyperparameter tuning.

5.2. Baselines

We evaluate the performance of our proposed ComRec model against several state-of-the-art baselines, which are categorized into three groups: (1) *Rule-Based Methods* including NCF-AVG, NCF-LM, and NCF-MAX; (2) *Preference-Based Methods* such as AGREE, GroupIM, CubeRec,

[1] https://www.yelp.com/dataset

[2] https://grouplens.org/datasets/movielens/

Algorithm 1 Constructing MovieLens-Simi and MovieLens-Rand datasets.

Input: User rating matrix $\mathbf{Y}_U$, PCC threshold $\tau = 0.27$, Group size n_g
Output: Two datasets: MovieLens-Simi and MovieLens-Rand

Initialize empty datasets: SimiGroups ← [], RandGroups ← []
for all user u_i in R **do**
    Compute PCC between u_i and all other users
    Find $n_g - 1$ users whose PCC with $u_i > \tau$
    if such users exist **then**
        Form a group G of n_g users with high mutual similarity
        Add G to SimiGroups
    end if
end for
while not enough random groups **do**
    Randomly sample n_g users to form a group G
    Add G to RandGroups
end while
for all group G in SimiGroups ∪ RandGroups **do**
    Find all items rated > 4 by every member in G
    Define such items as ground-truth items for group G
end for
return SimiGroups and RandGroups with ground-truth labels

ConsRec, and DHMAE; and (3) *Knowledge-Aware Methods* comprising KGIN-AVG and KGAG.

- **NCF-AVG**, **NCF-LM**, and **NCF-MAX** are variants of the NCF model (He et al., 2017), a classical method for individual recommendations. These models employ different rule-based aggregation strategies to capture group preferences: averaging (AVG), least misery (LM), and maximum satisfaction (MAX).
- **AGREE** (Cao et al., 2018) jointly trains user-item and group-item interactions, aggregating the group representation by weighting each group member based on the attention scores computed for the candidate items.
- **GroupIM** (Sankar et al., 2020) addresses the challenge of data sparsity by enhancing the alignment between group and individual member representations through mutual information maximization.
- **KGAG** (Deng et al., 2021) is an attention based method that leverages a graph convolutional network to model the structural relationships between items and users within the KG.
- **KGIN-AVG** (Wang et al., 2021) employs a GNN to recursively integrate relational sequences on the KG, capturing user intent. The user representations learned by KGIN are aggregated via average pooling strategy to form the group representation.
- **CubeRec** (Chen et al., 2022) leverages the geometric properties of hypercubes to represent groups in vector space and computes preference scores between group hypercubes and item points using an enhanced distance metric.
- **ConsRec** (Wu et al., 2023) integrates member-level, item-level, and group-level preferences using multi-view learning and employs a hypergraph neural network for member aggregation.
- **DHMAE** (Zhao et al., 2024) presents a novel hypergraph masked autoencoder that utilizes generative self-supervised techniques to generate self-supervised signals. Additionally, it tackles data noise by employing a degree-sensitive hypergraph masking strategy.

5.3. Evaluation metrics and implementations

Following (Chen et al., 2022; Wu et al., 2023; Zhao et al., 2024), we evaluate the performance of ComRec using two widely adopted metrics: Recall at rank K (R@K) and NDCG at rank K (N@K), where K = 10, 20. R@K measures the proportion of relevant items that are correctly retrieved within the top-K recommendations. It is defined as:

$$\mathrm{R@K} = \frac{|\mathrm{Rel}_u \cap \mathrm{Rec}_u^K|}{|\mathrm{Rel}_u|}, \tag{24}$$

where Rel_u denotes the set of relevant (i.e., truly preferred) items for user u, and Rec_u^K denotes the top-K items recommended to user u. In contrast, N@K evaluates not only the presence of relevant items in the recommendation list but also their positions, giving higher-ranked relevant items greater influence on the overall score. The metric is defined as:

$$\mathrm{NDCG@K} = \frac{\mathrm{DCG@K}}{\mathrm{IDCG@K}}, \tag{25}$$

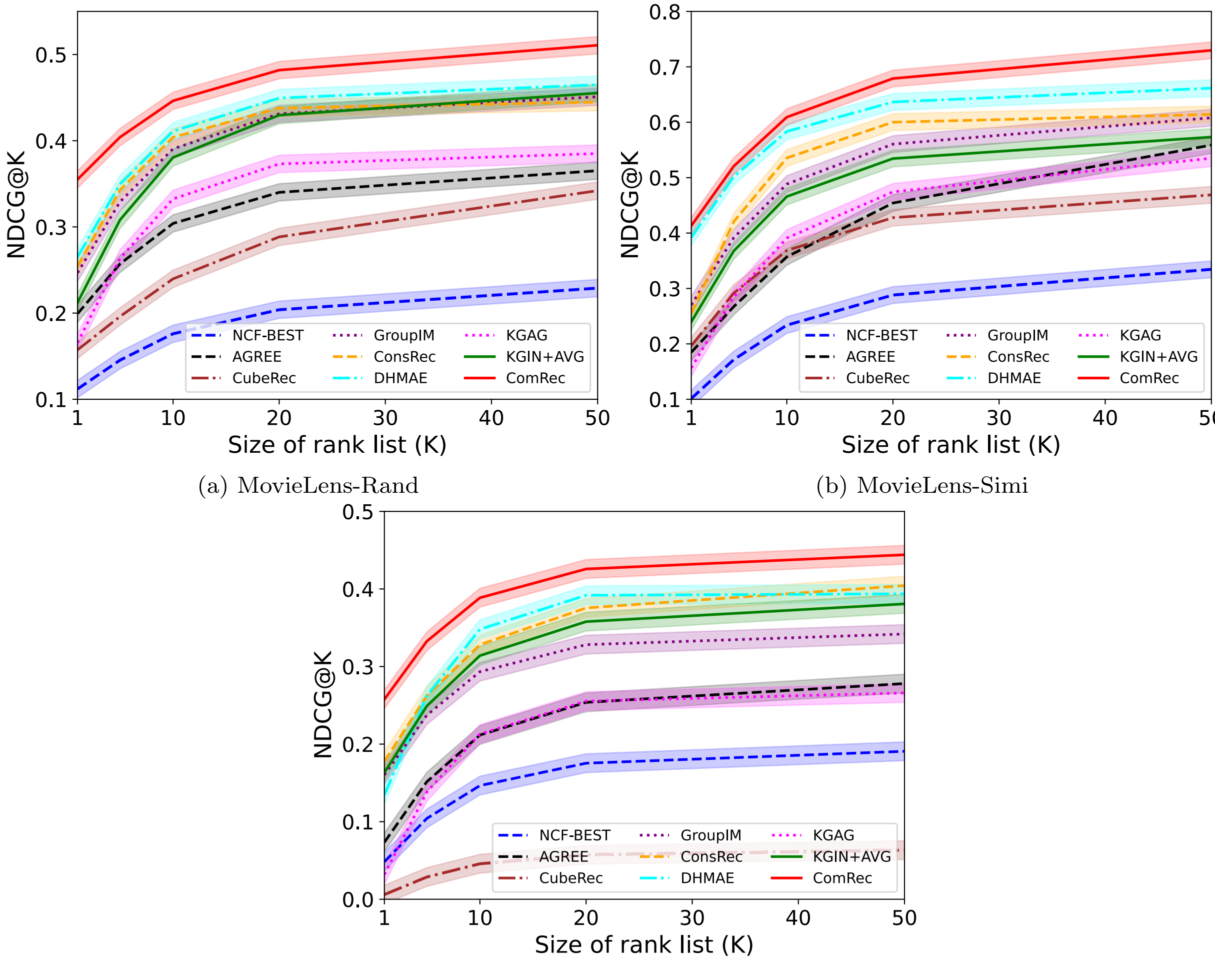
$$\mathrm{DCG@K} = \sum_{i=1}^{K} \frac{2^{rel_i} - 1}{\log_2(i+1)}, \tag{26}$$

where rel_i denotes the true relevance of the i-th recommended item. DCG@K applies a logarithmic discount to relevance scores based on the position, penalizing lower-ranked items more heavily. IDCG@K corresponds to the DCG@K score of the ideal (perfect) ranking, where all relevant items are ordered at the top according to their relevance.

Additionally, we introduce the Value Improvement Percentage (VIP) to quantify the performance enhancement of our method relative to the

Table 4
Overall performance comparison on three datasets. (Note: * denotes the statistical significance for p-value < 0.05 compared to the best baseline, the boldface indicates the best model result of the dataset, and the underline indicates the second best model result of the dataset.).

Dataset	Yelp				MovieLens-Rand				MovieLens-Simi			
Metric	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20
Rule-Based Methods												
NCF + AVG	0.1492	0.1560	0.2182	0.2571	0.1311	0.1633	0.3234	0.4113	0.1992	0.2511	0.3469	0.4339
NCF + LM	0.1016	0.1254	0.1771	0.1984	0.1734	0.2027	0.3761	0.4602	0.2463	0.3130	0.4004	0.5161
NCF + MAX	0.1025	0.1366	0.1899	0.2216	0.1389	0.1815	0.3452	0.4269	0.2081	0.2636	0.3626	0.4523
Preference-Based Methods												
AGREE	0.2071	0.2315	0.3649	0.4789	0.2907	0.3549	0.4428	0.6249	0.3468	0.4951	0.6638	0.7294
CubeRec	0.0478	0.0632	0.0571	0.1077	0.2416	0.2719	0.4335	0.5518	0.3823	0.4210	0.6437	0.7816
GroupIM	0.2868	0.3143	0.4008	0.4956	0.3977	0.4109	0.4311	0.6296	0.4637	0.5849	0.6718	0.7616
ConsRec	0.3218	0.3464	0.4155	0.5105	0.4069	0.4266	0.4692	0.6495	0.5379	0.6476	0.7117	0.8217
DHMAE	0.3529	0.3777	0.4371	0.5023	0.4108	0.4395	0.4670	0.6335	0.5422	0.6533	0.7390	0.8408
Knowledge-Aware Methods												
KGAG	0.2195	0.2451	0.3862	0.4975	0.3157	0.3788	0.4480	0.6387	0.3858	0.4863	0.6435	0.7476
KGIN + AVG	0.3017	0.3435	0.3343	0.4282	0.3879	0.4179	0.4306	0.6105	0.4571	0.5639	0.6636	0.7461
ComRec	**0.3831***	**0.3958***	**0.4697***	**0.5382***	**0.4289***	**0.4692***	**0.4932***	**0.6846***	**0.5704***	**0.6914***	**0.7728***	**0.8749***
VIP of ComRec	8.56 %	4.79 %	7.46 %	5.43 %	4.41 %	6.76 %	5.12 %	5.40 %	5.20 %	5.83 %	4.57 %	4.06 %



(a) MovieLens-Rand

(b) MovieLens-Simi

(c) Yelp

Fig. 3. NDCG@K across varying values of the rank list size K, with shaded variance bands representing 95 % confidence intervals calculated over 10 independent runs.

baseline models, defined as follows:

$$VIP = \frac{v_a - v_b}{v_b} \times 100\,\%, \tag{27}$$

where v_a represents the metric value of our model, and v_b denotes the metric value of the baseline model for comparison. We conducted 5-fold cross-validation on each dataset to minimize bias. Since the data do not follow a normal distribution, we applied the permutation test (Nichols & Holmes, 2002) for significance testing.

To ensure reproducibility, we provide the following hyperparameter settings. Specifically, the embedding size d is set to 64, the batch size to 1024, the number of graph neural network layers l to 3, the number of attention heads h to 8, and the number of group commonalities $|C|$ to 4. The learning rate, along with the coefficients for the additional constraints, λ_1 and λ_2, are tuned through a grid search. The learning rates are chosen from $\{1e^{-4}, 1e^{-3}, 1e^{-2}\}$, λ_1 from $\{0.001, 0.01, 0.1\}$, and λ_2 from $\{1e^{-6}, 1e^{-5}, 1e^{-4}, 1e^{-3}\}$. The Adam (Kingma & Ba, 2015) optimizer is used for ComRec, and for each ground truth item, 5 negative items are sampled. All experiments are carried out on a single Nvidia GeForce RTX 4090 GPU, using PyTorch implementations on a Linux platform.

5.4. Result and analysis

5.4.1. Overall performance comparison (RQ1)

The overall results of the comparison between ComRec and state-of-the-art baselines are summarized in Table 4. It is evident that ComRec consistently surpasses all baseline models across the three datasets, showing superior performance in both R@K and N@K metrics.

Specifically, ComRec achieves the greatest improvement over rule-based methods across all datasets in both metrics. For example, in terms of N@10, ComRec outperforms NCF-BEST[3] by 156.8 % on Yelp, 147.3 % on MovieLens-Rand, and 131.6 % on MovieLens-Simi. This improvement may stem from ComRec's data-driven strategy, which dynamically balances group commonality and diversity to generate more robust group representations. In contrast, rule-based predefined strategies overlook real interactions among group members and struggle to adapt to dynamic scenarios.

By comparison, while ComRec's advantages over preference-based methods are relatively smaller, the improvements remain significant, with a p-value < 0.05 according to the permutation test. As demonstrated in Table 4, ComRec consistently delivers superior performance over the strongest preference-based baseline, DHMAE, across all datasets. Specifically, for N@10, the VIP reaches 8.56 % on Yelp, 4.41 % on MovieLens-Rand, and 5.20 % on MovieLens-Simi; similarly, for R@10, the corresponding improvements are 7.46 %, 5.61 %, and 4.57 %, respectively. This is because ComRec goes beyond individual-level preference aggregation (group difference in this paper) by also

[3] NCF-BEST denotes the NCF-based method that achieves the best performance across different datasets, specifically, NCF + AVG for Yelp and NCF + LM for MovieLens.

leveraging group commonality information in the G-CKG to jointly model the group decision-making process.

It is worth noting that our model significantly outperforms other knowledge-aware methods in both N@K and R@K metrics. For example, on the Yelp dataset, N@10 improves by 27.0 % over KGIN + AVG, while the average improvement in N@10 across several other datasets is 18.3 %. Similarly, the average improvement in R@10 over KGIN + AVG across all datasets is 21.5 %. These results empirically validate that ComRecs use of G-CKG to model group-level commonality information is more effective than simply enhancing individual-level knowledge representations through KG. Additionally, the innovatively designed commonality attention mechanism is critical for accurately simulating group decision-making processes and providing better recommendation results.

In addition, when comparing performance across datasets, we observe that all models tend to yield relatively lower scores on the Yelp dataset than on MovieLens-Rand and MovieLens-Simi. As shown in Table 3, Yelp exhibits the highest data sparsity among the three datasets, with each group having an average of only 1.38 interactions, making it difficult to learn robust group representations. Despite this, the proposed approach consistently demonstrates significant improvements (e.g., an average VIP of 6.68 % for N@K and 6.45 % for R@K). This can be attributed to the fact that, unlike preference-based methods that focus on individual preferences, our model focuses on extracting consistent commonality information across groups, effectively alleviating the sparsity issue and maintaining a stable advantage even on such a challenging dataset. Another key observation from Table 4 is that all models tend to perform better on the MovieLens-Simi dataset than on MovieLens-Rand, as reflected in both N@K and R@K metrics. Specifically, the average N@10 and R@10 values for MovieLens-Simi are 0.395 and 0.602, whereas for MovieLens-Rand, these values drop to 0.302 and 0.424. This difference is expected, as MovieLens-Simi groups are composed of users with similar preferences, while the groups in MovieLens-Rand are randomly formed, lacking preference alignment.

To assess the effectiveness ComRec further, we perform an extensive comparison of NDCG scores across various cutoff values (K = [1, 50]) for all models. This evaluation metric is deliberately chosen, as NDCG emphasizes the ranking of positive instances more effectively than HR, thus allowing for a fairer evaluation of model performance. As shown in Fig. 3, ComRec consistently outperforms all baseline methods across the three datasets and under all ranking cutoff values (K = 1 to 50), demonstrating its superior ranking capability. The advantage is especially evident at smaller K values (e.g., @1, @5, @10), highlighting ComRec's effectiveness in prioritizing the most relevant items at the top ranks. Notably, on the MovieLens-Simi dataset, ComRec performs comparably with DHMAE at smaller cutoff values (e.g., K = 1), which suggests that both models are effective in identifying the most representative items when user preferences within the group are highly aligned. However, as K increases, ComRec begins to outperform DHMAE by a clear margin, indicating its stronger ability to move beyond top-1 or top-5 recommendations and maintain relevance across deeper ranks. On the MovieLens-Rand and Yelp datasets-where group diversity and data sparsity are more pronounced-ComRec demonstrates clear and consistent improvements over all baselines, including DHMAE and ConsRec. These results validate the robustness and generalizability of ComRec, especially in complex recommendation scenarios involving diverse user intents.

5.4.2. *Ablation study (RQ2)*

We conduct ablation studies on several model variants to assess the contribution of ComRec's key components. Table 5 summarizes the performance of all variants across datasets, evaluated using N@K and R@K at cutoff values of 10 and 20. We remove or replace different components of ComRec, as described in Section 4, and introduce four variants below: (1) **w/.TransH** replaces the GNN framework described in Section 4.1.2 with TransH (Wang, Zhang, Feng, & Chen, 2014) for encoding entities and relations; (2) **w/o.Com** disables the group commonality component by setting $\alpha = 0$ in Eq. (20), and instead uses only the group difference representation for final group modeling; (3) **w/o.Diff** eliminates the user-based difference fine-tuning module by setting β to 0 in Eq. (20), relying solely on the group commonality representation as the final group representation; (4) **w/o.Orth** sets λ_1 in Eq. (23) to 0, discarding the orthogonal constraint $\mathcal{L}_{orth}$ on group commonality representations.

Table 5
The comparison between ComRec and its variants.

Dataset	Metric	w/.TransH	w/o.Com	w/o.Diff	w/o.Orth	ComRec
Yelp	N@10	0.1635	0.2440	0.3105	0.3588	**0.3831***
	N@20	0.1963	0.2665	0.3367	0.3643	**0.3958***
	R@10	0.1984	0.2936	0.4198	0.4359	**0.4697***
	R@20	0.2585	0.3634	0.4870	0.5104	**0.5382***
MovieLens-Rand	N@10	0.2251	0.3256	0.3641	0.4078	**0.4289***
	N@20	0.2869	0.3582	0.3984	0.4257	**0.4692***
	R@10	0.2336	0.3618	0.4248	0.4662	**0.4932***
	R@20	0.2923	0.4811	0.5872	0.6461	**0.6846***
MovieLens-Simi	N@10	0.2697	0.3972	0.4712	0.5184	**0.5704***
	N@20	0.4158	0.5205	0.6010	0.6532	**0.6914***
	R@10	0.3771	0.5724	0.6945	0.7351	**0.7728***
	R@20	0.4465	0.7169	0.7789	0.8268	**0.8749***

It is evident from Table 5 that **ComRec** consistently outperforms all four ablated variants across the datasets, highlighting the necessity of each model component. Since the trends are largely consistent, we take the Yelp dataset as a representative case to discuss the main findings below:

ComRec demonstrates the most significant improvement over **w/.TransH**, with VIP values of N@10 at 134.3 % and R@10 at 136.7 %. This is understandable, as **w/.TransH** utilizes TransH to encode knowledge graph nodes by focusing solely on local triple-level information, while neglecting the high-order structural relationships among nodes. Consequently, it is unable to effectively capture group-level commonality and difference.

The performance degradation from removing the commonality modeling module in ComRec is noticeably greater than that caused by removing the difference fine-tuning module. Specifically, in **w/o.Com**, the VIP of N@10 and R@10 decreases by 57.0 % and 60.0 %, respectively, whereas **w/o.Diff** results in smaller reductions of 23.4 % for N@10 and 11.9 % for R@10. It indicates that group commonality plays a more essential role in modeling the group decision-making process, while group difference information serves a supportive role, helping to refine the representation of group commonality. This is likely because the group commonality module learns general decision-making patterns from the G-CKG, providing the foundation for the final group representation. While the group difference module can emphasize the diverse preferences of different members using multi-head attention, it struggles to learn robust group representations from sparse group-item interaction data-an issue that has long challenged attention-based methods. However, the effective combination of both modules helps address these limitations.

The orthogonal constraint, while contributing the least to performance improvement among all modules in ComRec, still provides a notable gain. For example, in the case of **w/o.Orth**, the VIP for N@10 increases by 6.8 % and for R@20 by 7.8 %. It aligns with expectations, as enforcing independence among group commonality representations not only improves the interpretability of the group decision-making process but also decouples the key factors that influence group decisions. In essence, it promotes more informative and disentangled group representations, which allows the model to better capture diverse decision-making patterns and ultimately leads to improved recommendation accuracy.

Table 6
Performance comparison of ComRec and ConsRec with swapped group commonality modeling methods.

Dataset	Metric	ConsRec	w/.G-CKG	ComRec	w/.Overlap
Yelp	N@10	0.3218	<u>0.3347</u> (+4.0 %)	**0.3831***	0.2723 (-28.9 %)
	N@20	0.3464	<u>0.3595</u> (+3.8 %)	**0.3958***	0.3087 (-22.0 %)
	R@10	0.4155	<u>0.4289</u> (+3.2 %)	**0.4697***	0.3245 (-30.9 %)
	R@20	0.5105	<u>0.5199</u> (+1.8 %)	**0.5382***	0.4062 (-24.5 %)
MovieLens-Rand	N@10	0.4069	<u>0.4194</u> (+3.1 %)	**0.4289***	0.3371 (-21.4 %)
	N@20	0.4266	<u>0.4472</u> (+4.8 %)	**0.4692***	0.3766 (-19.7 %)
	R@10	0.4692	<u>0.4838</u> (+3.1 %)	**0.4932***	0.3894 (-21.0 %)
	R@20	0.6495	<u>0.6646</u> (+2.3 %)	**0.6846***	0.5158 (-24.7 %)
MovieLens-Simi	N@10	0.5379	<u>0.5496</u> (+2.2 %)	**0.5704***	0.4119 (-27.8 %)
	N@20	0.6476	<u>0.6705</u> (+3.5 %)	**0.6914***	0.5511 (-20.3 %)
	R@10	0.7117	<u>0.7231</u> (+1.6 %)	**0.7728***	0.6167 (-20.2 %)
	R@20	0.8217	<u>0.8389</u> (+2.1 %)	**0.8749***	0.7496 (-14.3 %)

5.4.3. Effectiveness of group commonality (RQ3)

To verify whether the innovative design of ComRec, which mines fine-grained group commonality from G-CKG, is more effective than the simpler approach of using an overlap graph to propagate group similarity based on shared members for indirect modeling, we swapped the group commonality modeling methods between ComRec and ConsRec. Specifically, we replaced the group-level embeddings $\bar{\mathbf{G}}^g$ from ConsRec's Section 4.4 (Wu et al., 2023) with $\mathbf{e}_{g_{com}}$ obtained using the method described in Section 4.2 of this paper, resulting in the variant **w/.G-CKG**; similarly, we replaced ComRec's $\mathbf{e}_{g_{com}}$ with $\bar{\mathbf{G}}^g$ from ConsRec to obtain the variant **w/.Overlap**. Table 6 presents a performance comparison between ConsRec, ComRec, and their variants across all datasets, using N@K and R@K (K = 10, 20) as evaluation metrics.

As shown in the table, replacing G-CKG with the overlap graph leads to a consistent performance decline for the **w/.Overlap** variant compared to ComRec across all datasets and metrics, with noticeable drops observed. For instance, on the Yelp dataset, the N@10 score decreases by 28.9 %, while the R@10 score on the MovieLens-Rand dataset drops by 21.0 %. These results highlight the limitations of the overlap graph approach, which relies on coarse edge weights derived from member overlap and fails to effectively capture shared behavioral patterns among groups.

In contrast, the **w/.G-CKG** variant consistently outperforms the original ConsRec across all three datasets, achieving an average improvement of 3.1 % in N@10 and 2.63 % in R@10. This demonstrates that modeling group commonality from G-CKG and aggregating it via a commonality-aware attention mechanism enables more accurate representation of group-level shared characteristics. Moreover, this module proves to be not only effective within ComRec but also easily transferable to other models, yielding noticeable performance gains. We believe that further tailoring group commonality representations to specific model architectures could unlock even greater improvements.

5.4.4. Hyperparameter analysis (RQ4)

To explore RQ3, we evaluate the effects of the latent dimension d, the number of commonalities $|\mathcal{C}|$, the orthogonal loss control parameter λ_1, and the L_2 regularization parameter λ_2 across all datasets. While tuning d and $|\mathcal{C}|$, we keep the other hyperparameters fixed, and the results are presented in Fig. 4. We also conduct a grid search to identify the optimal combinations of λ_1 and λ_2, with the results shown in the heatmap in Fig. 5. N@20 and R@20 are used as the evaluation metrics.

Impact of d. We tune d in $\{16, 32, 64, 128\}$. It is observed that the model's performance improves initially as d increases, since larger

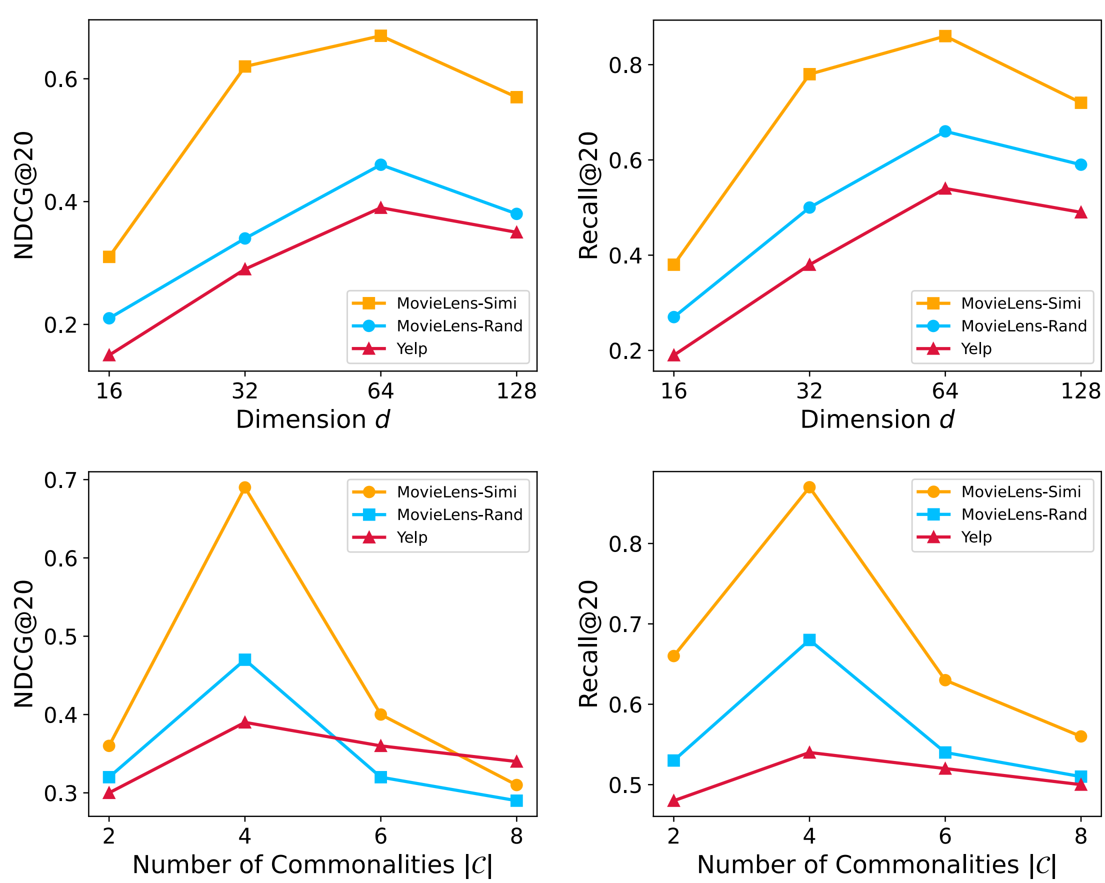


Fig. 4. Performance of ComRec for different dimensions d and number of commonalities $|\mathcal{C}|$.

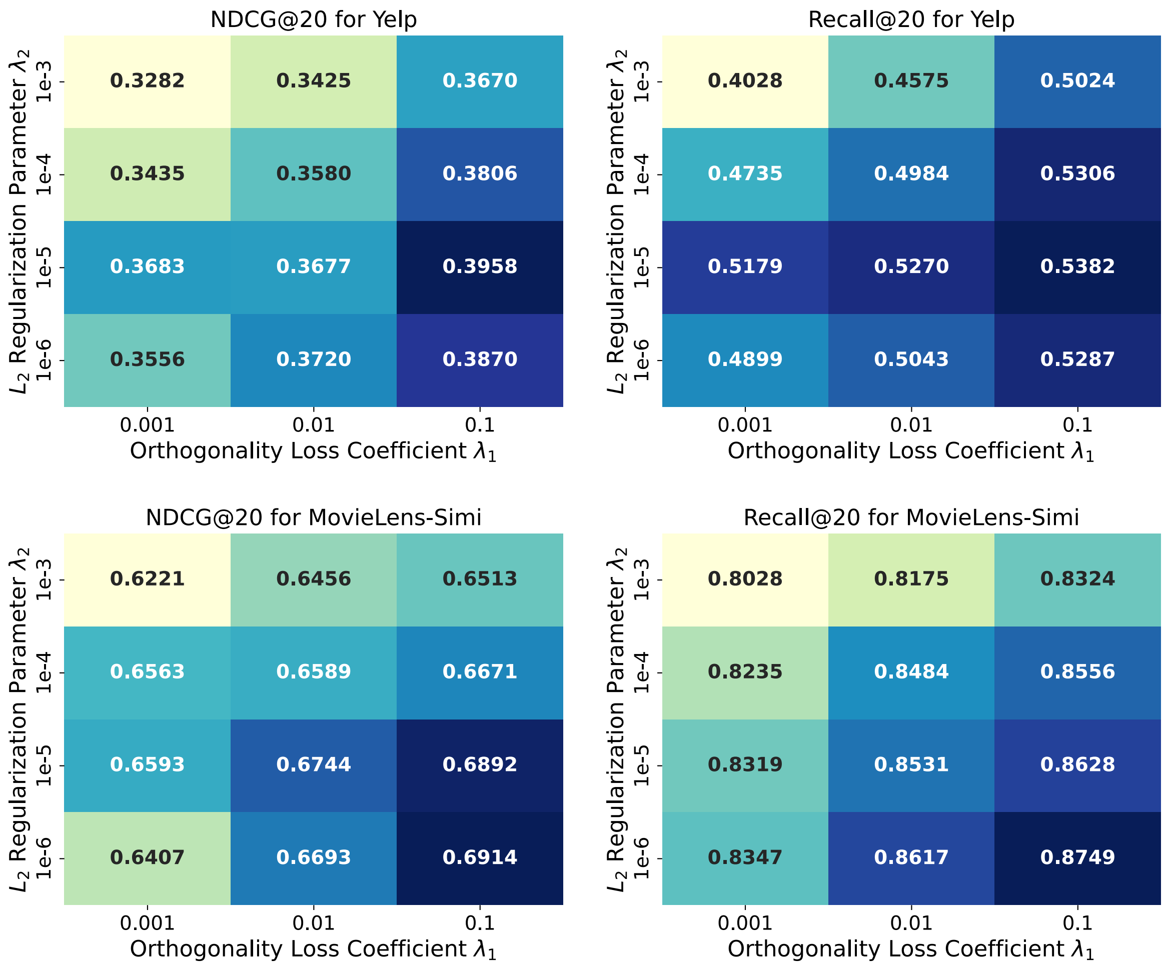


Fig. 5. Performance of ComRec for different hyperparameter pairs (λ_1, λ_2). Darker colors indicate better recommendation performance.

embeddings can capture more meaningful information. However, once d reaches a certain threshold (128 in our case), the sparsity of the data may cause overfitting, leading to a decline in performance.

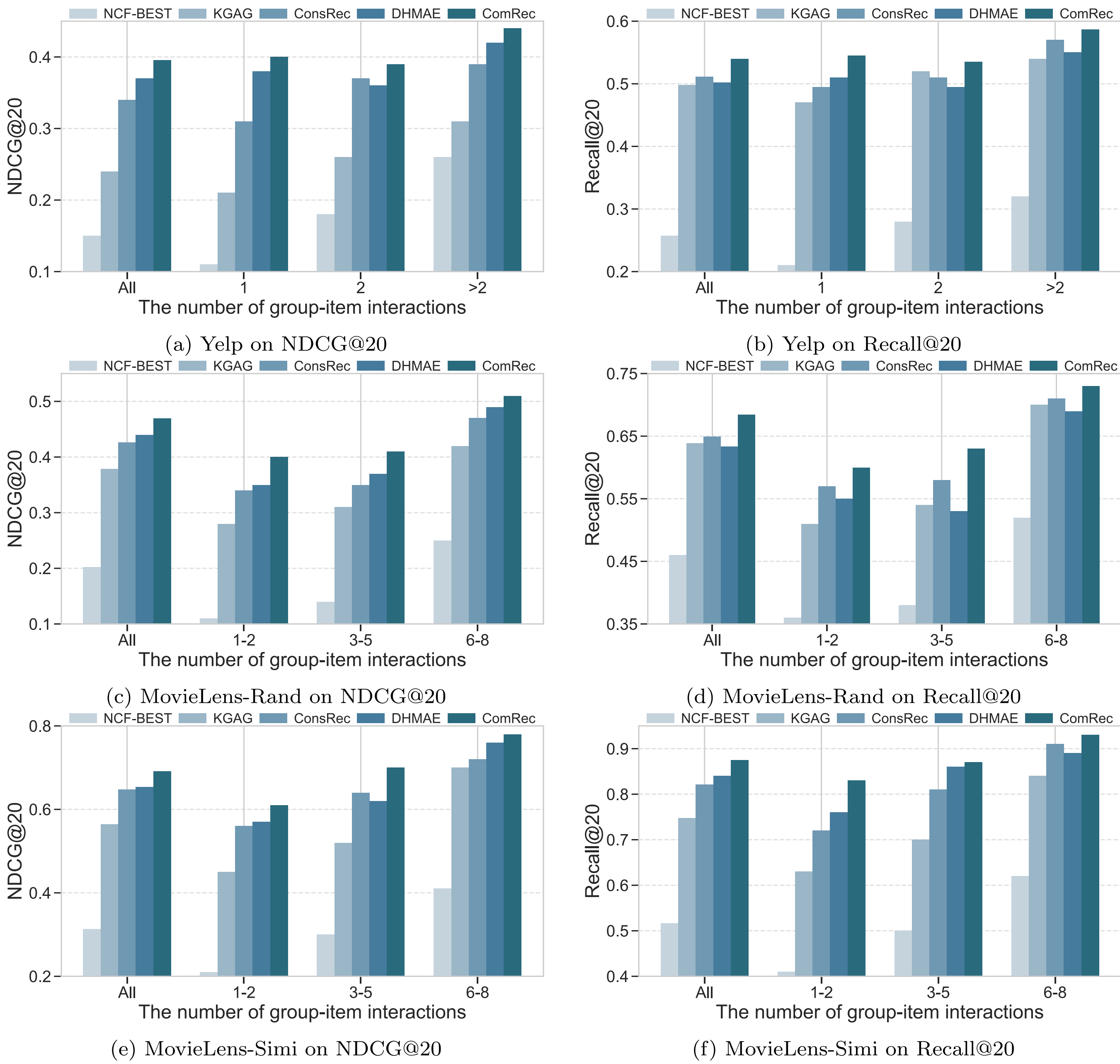
Impact of $|C|$. We test the sensitivity of commonality by varying their number within the set $\{2, 4, 6, 8\}$. Overall, the model's performance shows an initial increase followed by a decline as the number of commonalities grows. This can be explained by the fact that when the number of commonalities is small, the captured group commonalities are limited, leading to reduced model effectiveness. However, when there are too many commonalities, including irrelevant relationships, noise is introduced, negatively affecting the model's performance. We believe this discrepancy stems from differences in the relationship sets within the KGs of the two datasets. Specifically, the Yelp dataset incorporates attributes such as public ratings and noise levels, providing a more diverse set of relationships. In contrast, the MovieLens-Simi dataset relies on manually generated relationships with limited behaviors, resulting in a less diverse relationship set and potentially more noise in the data.

Impact of (λ_1, λ_2). We perform a grid search to investigate the impact of the orthogonality loss coefficient λ_1 and the L_2 regularization parameter λ_2 on model performance. Fig. 5 presents the experimental results on the Yelp and MovieLens-Simi datasets in terms of NDCG@20 and Recall@20. On the Yelp dataset, the best NDCG@20 score (0.3958) is achieved when $\lambda_1 = 0.1$ and $\lambda_2 = 1e^{-5}$. Under the same setting, the model also attains the highest Recall@20 score (0.5382). This suggests that a moderate orthogonality constraint effectively enhances the independence among group commonality embeddings, thereby improving the discriminative ability of the commonality attention mechanism during member information aggregation. Notably, when λ_2 is set to a large value (e.g., $1e^{-3}$), the model performance degrades noticeably, indicating that overly strong regularization may hinder the model's ability to capture user preferences. A similar trend is observed on the MovieLens-Simi dataset. When $\lambda_1 = 0.1$ and $\lambda_2 = 1e^{-6}$, the model achieves the best results on both NDCG@20 (0.6914) and Recall@20 (0.8749). This trend further validates that a stronger orthogonality constraint paired with a minimal regularization term yields optimal performance, highlighting the positive role of maintaining independence among commonality embeddings in improving member aggregation quality and overall recommendation effectiveness in GR tasks.

5.4.5. Effect of the number of group-item interactions (RQ5)

To better understand how the number of group-item interactions affects recommendation performance, especially in sparse interaction scenarios, we apply the grouping strategy outlined in Yang et al. (2024). Specifically, the interaction data of the MovieLens-Simi and MovieLens-Rand datasets are divided into three groups: 1-2, 3-5, and 6-8 interactions. For the Yelp dataset, due to its inherently high sparsity, we categorize the data into 1, 2, and more than 2 interactions. Fig. 6 presents the N@20 and R@20 scores of ComRec and four baseline models (NCF-BSET, KGAG, ConsRec, and DHMAE) across all datasets.

The results demonstrate that ComRec exhibits strong robustness across different levels of group-item interactions, consistently outperforming all baseline models on the three datasets. This trend aligns well with the overall performance observed in Table 4. Notably, ComRec shows a clear advantage over other baseline models in extremely sparse interaction settings (e.g., 1-2 interactions in Yelp, and 1-5 in MovieLens). This includes models specifically designed to address data sparsity, such



(a) Yelp on NDCG@20

(b) Yelp on Recall@20

(c) MovieLens-Rand on NDCG@20

(d) MovieLens-Rand on Recall@20

(e) MovieLens-Simi on NDCG@20

(f) MovieLens-Simi on Recall@20

Fig. 6. Performance of ComRec and baseline models across different numbers of group-item interactions.

as DHMAE, which mitigates sparsity via self-supervised signals from a hypergraph masked autoencoder, and KGAG, which leverages KG to enhance user representations. While DHMAE enhances group representations through structural completion at the individual member level, ComRec focuses on learning member-independent general group commonalities at the group level first, then combines them with specific member compositions. This combination of representations at both the group and user levels leads to more robust group representations, particularly in sparse data scenarios. For KGAG, our robustness advantage lies in the full utilization of the KG. While simply incorporating the KG as external information can partially alleviate group data sparsity (as seen in Fig. 6(b) when the number of interactions is 2), our specialized design based on group commonalities enables a deeper integration with the KG, allowing for better modeling of group-level patterns and more effective handling of sparse interactions.

In addition, some interesting observations emerge from the Yelp dataset. Our model and DHMAE show a slight decline in performance when the number of group interactions is 2, compared to when there is only a single interaction. As presented in Table 3, the Yelp dataset contains a considerable quantity of candidate items. The introduction of a second interaction may bring in two items that are far apart in the KG, introducing more noise and affecting the model's performance. In contrast, models like NCF-BEST and ConsRec, which lack additional constraints, are better able to mitigate the noise impact in the same scenario.

5.4.6. *Interpretability of ComRec (RQ6)*

To gain a deeper understanding of how group commonality and difference influence group decision-making, and to provide an intuitive insight of the interpretability of our model, we conduct a case study based on a real group instance from the Yelp dataset. This study aims to analyze how the group decision is inferred by tracing back the weights associated with the recommended item. Specifically, as shown in Fig. 7, the final representation of group g_{17199} consists of a

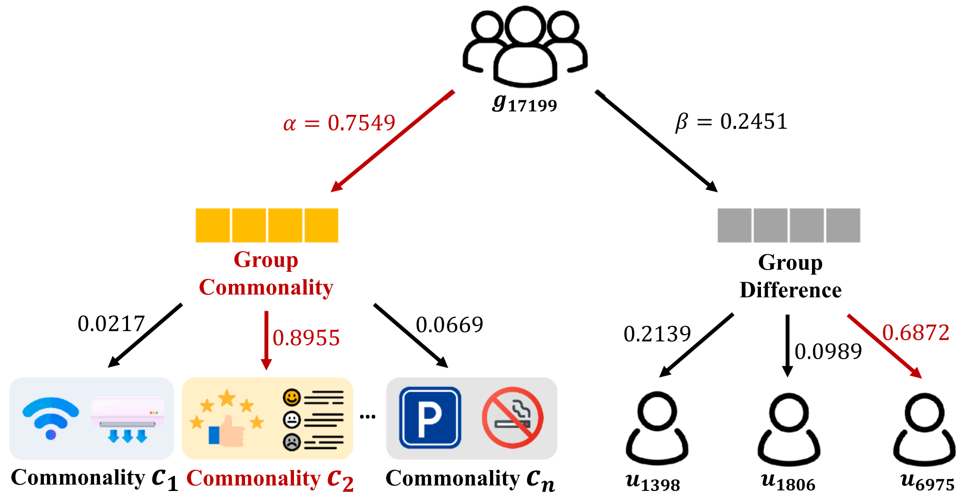
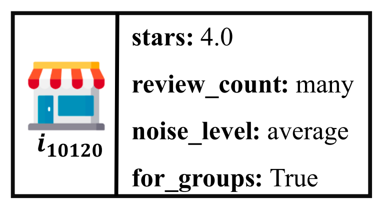


Fig. 7. A real case from the Yelp dataset demonstrating how ComRec generates interpretable recommendations. The identified group commonalities, shown in the bottom left corner, include: c_1: Wi-Fi and air conditioning; c_2: positive reviews and a high review count; c_n: parking availability and a no-smoking policy.

Relations	Ratios
r_2 - stars	47.21%
r_{27} - review_count	25.44%
r_{13} - noise_level	7.09%

Top-3 contributing KG relations in c_2



KG attributes of interacted item i_{10120}

Fig. 8. The contribution proportions of different relations that form commonality c_2 and the KG attributes of item i_{10120} interacted with by group g_{17199}.

commonality representation weighted at 0.7549 and a difference representation weighted at 0.2451, highlighting the dominant role of commonality in the group's decision-making process, which aligns with the findings of our ablation study in Section 5.4.2.

To further understand how this representation is constructed, we trace back the process by which the commonality attention mechanism derives the commonality representation for group g_{17199}. Our goal is to identify which commonality embeddings contribute most and which specific relations in the relation set $\mathcal{R}'$ give rise to these embeddings. We find that the most influential commonality embedding is c_2, which is primarily derived from two business-related relations: star rating and review count. In the group difference representation used to fine-tune the group's overall representation, we identify three users with differing levels of influence on the final decision. User u_{6975} is recognized as the primary decision-maker, carrying a weight of 0.6872, while user u_{1806} contributes only 0.0989. This is reasonable, as individuals within a group hold varying degrees of influence, which our multi-head attention mechanism successfully captures.

After understanding the composition of the group representation, we further explore how the group commonality representation impacts the final decision. In Fig. 8, we present the top-3 relations that form commonality c_2 for group g_{17199}, along with the relevant attributes of item i_{10120}. We observe that i_{10120}'s high rating (4.0 out of 5.0) and large review count align with the preferences of group g_{17199}, indicating that the group's choice of i_{10120} is primarily influenced by its high rating, a large number of reviews, and an acceptable noise level.

6. Discussion

6.1. Overview of experimental results

In conclusion, the comprehensive performance comparison results (Table 4) demonstrate that the proposed ComRec significantly outperforms existing preference-based methods (Cao et al., 2018; Chen et al., 2022; Sankar et al., 2020; Wu et al., 2023; Zhao et al., 2024) in terms of top-k recommendation metrics. This superiority is further supported by the detailed NDCG@K comparison (Fig. 3), which consistently shows improved ranking performance across different K values. Moreover, the ablation results (Table 5) confirm the critical role of group commonality in modeling group decision-making, with group difference information serving as a complementary factor that helps refine the commonality representation.

Beyond the overall results, this study offers several practical implications for GR. Firstly, the proposed Group-based Commonality Modeling has been demonstrated to seamlessly integrate with existing GR models and consistently enhance their performance (Table 6). Unlike previous approaches (Jia et al., 2021; Wu et al., 2023; Xu et al., 2024) that primarily rely on the number of overlapping members to model group commonality, our method not only incorporates such member overlap relations by constructing the G-CKG, but also captures fine-grained commonality from the item side. This design effectively overcomes the limitations of prior methods, which tend to fail when there are no shared members across groups.

Another important contribution of ComRec is that it provides a novel perspective for addressing the data sparsity problem in GR, outperforming existing SSL-based methods (Li et al., 2023; Zhang et al., 2021; Zhao et al., 2024) (Fig. 6). ComRec models the group decision-making process in two stages: first, shared behavioral patterns set the general decision direction; then, individual member preferences refine the final choice. This avoids the disruptions caused by random member supplementation and preserves group structure, leading to better performance under sparse data conditions.

Finally, this study provides new perspectives for enhancing the interpretability of GR systems. Specifically, the case analysis (Fig. 7) reveals the internal structure of group decision-making. The results show that commonality information dominates the decision process, while individual members contribute differently to the differential signals, reflecting varied influence levels consistent with real-world dynamics. Moreover, ComRec identifies meaningful alignments between item attributes and group-level commonality representations (Fig. 8), offering practical guidance for developing interpretable GR applications.

6.2. Limitations

From a dataset perspective, the current approach benefits from extracting group commonality information from the KG. However, in practical applications, KGs may lack sufficient relationships and entities, particularly in specialized domains. Furthermore, KGs typically need to be continuously updated to incorporate new entities and relationships over time. Therefore, leveraging the powerful capabilities of large language models to automate the creation of dynamically updated KG (Ye, Zhang, Chen, & Chen, 2022; Zhang & Soh, 2024; Zhong, Wu, Li, Peng, & Wu, 2023) will be highly beneficial in future research.

From a model design perspective, although we infer similar behavioral models among groups based on group commonality information, the hyperparameter $|C|$ controls the number of group commonalities, while the hyperparameter λ_1 controls the strength of the orthogonal constraints between group commonalities. The optimal values for $|C|$ and λ_1 vary across different datasets and applications. An alternative solution is to leverage meta-learning techniques (Hospedales, Antoniou, Micaelli, & Storkey, 2021; Ren, Zeng, Yang, & Urtasun, 2018) to automate the elimination of such hyperparameters.

7. Conclusions

Existing preference-based models either overlook group commonality or rely on coarse-grained similarity measures based on the number of shared users between groups. This study introduces a new approach: Knowledge-Aware Modeling of Group Commonality for Group Recommendation (ComRec). ComRec leverages a G-CKG to extract fine-grained representations of group commonality, with orthogonal constraints imposed to ensure their independence. A commonality-aware attention mechanism is employed to aggregate group members based on shared commonalities, followed by a multi-head attention module to capture member-level differences within the group.

We evaluate our model on three real-world datasets (i.e., Yelp and MovieLens-20M), and the experimental results demonstrate the following: (1) our model outperforms state-of-the-art baselines in GR tasks; (2) group commonality plays a crucial role in the group decision-making process, while group difference serve to fine-tune the representation of group commonality; (3) fine-grained modeling of group commonality from the G-CKG is more effective than coarse-grained similarity propagation based on the number of shared users in overlap graphs; and (4) the combination of group-level commonality information with user-level difference information not only alleviates group data sparsity but also enhances the interpretability of GR results.

In the future, on one hand, we aim to tackle the limitations of the current approach, as detailed in Section 6.2. On the other hand, we plan to expand the application of ComRec to cross-domain GR scenarios. Data sparsity has long been a challenge in GR, but group commonality information can act as a bridge between domains, offering transferable group decision-making behaviors and patterns. This approach enables ComRec to effectively mitigate the issue of group data sparsity by leveraging shared group commonality information across different domains.

CRediT authorship contribution statement

Wen Wu: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization; **Guangze Ye:** Conceptualization, Methodology, Formal analysis, Investigation, Writing – original draft, Writing – review & editing, Supervision; **Hui Yu:** Writing – review & editing; **Wenxin Hu:** Writing – review & editing; **Xi Chen:** Writing – review & editing; **Liang He:** Writing – review & editing, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Wen Wu reports financial support was provided by National Natural Science Foundation of China. Xi Chen reports financial support was provided by The Research Project of Shanghai Science and Technology Commission. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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